

Computer Science XI

Exam revision cards — definitions, tables, mnemonics and ★
Golden topics on one glance each.

Bilingual support: English + اردو

امتحان چیت شیٹ — تعریفیں، جدول، یادداشت اور گولڈن موضوعات

01 Computer Systems

02 Computational Thinking & Algorithm

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Digital Literacy

XI

Contents

45 Golden cards are listed below. Start there if you have less than two hours.

Chapter 1: Computer Systems — 20 cards

Chapter 2: Computational Thinking & Algorithm — 13 cards

Chapter 3: Programming Fundamentals — 24 cards

Chapter 4: Data and Analysis — 15 cards

Chapter 5: Application and Impacts of Computing — 13 cards

Chapter 6: Digital Literacy — 17 cards

Night-before recaps: one-page summaries of every chapter

★ Golden index — if you only have two hours

These are the high-yield topics. Say each definition aloud, then draw the table or diagram from memory.

صرف دو گھنٹے ہوں تو یہ گولڈن کارڈز زبانی دہرائیں، پھر جدول حافظے سے بنائیں۔

1. ★ **Ch 1.** Analog and Digital Signals
2. ★ **Ch 1.** Logic Gates (Basic Gates: AND, OR, NOT)
3. ★ **Ch 1.** Logic Diagrams
4. ★ **Ch 1.** Karnaugh Maps (K-Maps)
5. ★ **Ch 1.** Software Development Life Cycle (SDLC)
6. ★ **Ch 1.** The Waterfall Model
7. ★ **Ch 1.** The Agile Model
8. ★ **Ch 1.** Communication Models and the OSI 7-Layer Model
9. ★ **Ch 2.** Algorithm
10. ★ **Ch 2.** Algorithm vs Pseudocode
11. ★ **Ch 2.** Decomposition
12. ★ **Ch 2.** Sorting Algorithms: Bubble Sort
13. ★ **Ch 2.** Sorting Algorithms: Selection Sort
14. ★ **Ch 2.** Searching Algorithms: Binary Search
15. ★ **Ch 3.** Fundamentals of Python Programming
16. ★ **Ch 3.** Variables
17. ★ **Ch 3.** Data Types in Python
18. ★ **Ch 3.** Operators and Operands: Arithmetic and Assignment
19. ★ **Ch 3.** Relational, Logical and Membership Operators
20. ★ **Ch 3.** Bitwise Operators
21. ★ **Ch 3.** Control Structures
22. ★ **Ch 3.** Selection Statements: if and if-else
23. ★ **Ch 3.** Selection Statements: if-elif-else and Nested if
24. ★ **Ch 3.** Repetition Statements: the for Loop and range()
25. ★ **Ch 3.** The while Loop, break and continue
26. ★ **Ch 3.** Nested Loops and for vs while
27. ★ **Ch 4.** Keys and Integrity Constraints
28. ★ **Ch 4.** Entity Relationship Model (ER-Model)
29. ★ **Ch 4.** Referential Integrity
30. ★ **Ch 4.** Creating Queries in MS Access
31. ★ **Ch 5.** Artificial Intelligence (AI)
32. ★ **Ch 5.** Internet of Things (IoT) & Components
33. ★ **Ch 5.** Information Sources (Primary, Secondary, Tertiary)
34. ★ **Ch 5.** Assistive Technologies
35. ★ **Ch 6.** Types of Data
36. ★ **Ch 6.** Data-Collection Strategies
37. ★ **Ch 6.** Interviews and Surveys
38. ★ **Ch 6.** Prototypes, Observation and Simulations
39. ★ **Ch 6.** Primary and Secondary Data
40. ★ **Ch 6.** Case Study: Digital Inquiry Project
41. ★ **Ch 6.** Steps 1 – 3: Advanced Search and Methodology
42. ★ **Ch 6.** Steps 4 – 6: Survey Preparation and Primary Data Collection
43. ★ **Ch 6.** Steps 7 – 9: Secondary Data and Spreadsheet Organisation
44. ★ **Ch 6.** Steps 10 – 12: Data Analysis, Conclusion and Recommendations
45. ★ **Ch 6.** The Final Digital Artefact

How to use these Cheat Sheets

This booklet is the **night-before and last-week** revision pack. Each topic is one card: the definition you must write, the table or formula, a memory trick, the usual exam wording, and one Urdu line. It is not a textbook. Open the Student's Edition when you need a worked example.

یہ چیٹ شیٹ امتحان سے پہلے رات اور آخری ہفتے کے لیے ہے۔ ہر موضوع ایک کارڈ ہے۔ مثالیں اصل کتاب میں ہیں۔

How to revise with this booklet

1. **Two hours left.** Read only the ★ Golden cards and the Golden index. Say each definition aloud.
2. **One day left.** One chapter at a sitting. Cover the card, write the table from memory, then check.
3. **The night before.** Read the night-before recaps. Do not start a new chapter.
4. **In the hall.** Remember the recipe: *Define. Explain. Example. Diagram. Working.*

What each card holds

- **Definition** — the 1-mark sentence. Write it exactly.
- **Points / table / snippet** — the 3-mark and 5-mark body.
- **Remember** — a mnemonic or trap to avoid.
- **Exam** — how the board asks it.
- **اردو** — the same idea in one spoken line.

★ **Golden** cards are high-yield. If time is short, skip non-Golden cards first, not these.

CHAPTER 1

Computer Systems

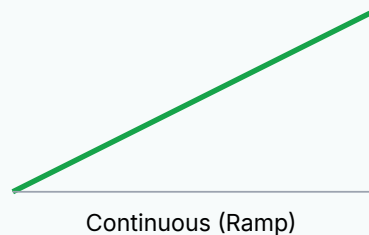
★ 8 Golden · 20 cards · Analog and Digital Signals · Logic Gates (Basic Gates: AND, OR, NOT) · Logic Diagrams · Karnaugh Maps (K-Maps) · Software Development Life Cycle (SDLC) · The Waterfall Model · The Agile Model · Communication Models and the OSI 7-Layer Model

1.1.1 Discrete and Continuous Quantities

Everything around you that can be measured or counted is a quantity.

- Quantities come in two kinds.
- Discrete quantities are separate, countable items with specific values.
- You can count them one by one: 1, 2, 3 ... There is nothing "in between".

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

Remember: Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask...

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔

1.1.2 Introduction to Digital Systems

A digital system is an electronic system that uses only two values: 0 and 1.

- 0 represents OFF, LOW, or FALSE.
- 1 represents ON, HIGH, or TRUE.

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Remember: Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

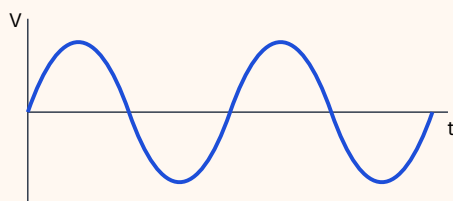
ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔

1.1.3 Analog and Digital Signals ★ Golden

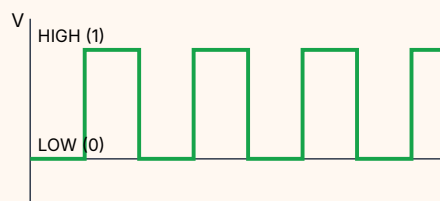
A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

اینا لاگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.4 Boolean Algebra

Boolean algebra was developed by the English mathematician George Boole.

- It is a kind of algebra that deals with only two values: TRUE (1) and FALSE (0).
- It is the foundation of digital electronics and computer programming — every if condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, Y = A + B

Trap: In Boolean algebra "+" means OR, not addition.

بولین الجبرا جارج بول (George Boole) نے بنایا۔

1.1.5 Boolean Operations

There are three basic Boolean operations. Every other operation can be made from them.

- Every other operation can be made from them.
- AND operation ($\cdot$) — gives output 1 only when ALL inputs are 1.
- Expression: $Y = A \cdot B$ (often written just AB).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

Remember: AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ).

1.1.6 Truth Tables

A truth table shows all possible input combinations and the output for each one, for a logic gate or a circuit.

- Count the number of inputs (n).
- Calculate total rows = 2^n .
- List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
- Calculate the output for each combination.

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

Remember: Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write...

ٹروٹھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

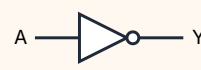
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

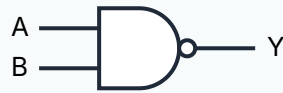
لاچک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.7 Universal Logic Gates (NAND and NOR)

NAND means "NOT AND": it is an AND gate followed by a NOT.

- NOR means "NOT OR": an OR gate followed by a NOT.
- On the symbol, the NOT is shown as a small bubble at the output.
- NAND and NOR are called universal gates because any logic circuit in the world can be built using only NAND gates (or only NOR gates).

A	B	A · B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

Remember: NAND is the exact opposite of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR.

1.1.7 Advanced Logic Gates (XOR and XNOR)

XOR (Exclusive OR) gives output 1 when the inputs are different.

- In terms of basic operations, $A \oplus B = A'B + AB'$.
- XNOR (Exclusive NOR) is XOR followed by NOT.
- It gives output 1 when the inputs are the same.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

Remember: Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the same. XNOR is sometimes called the "equality gate".

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

A logic expression (Boolean expression) combines variables with operations, for example $Y = A + B'C$.

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

- To evaluate it correctly you must follow the order of precedence, just like BODMAS in maths: 1.

A	B	C	B'	B'·C	Y = A + B'·C
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

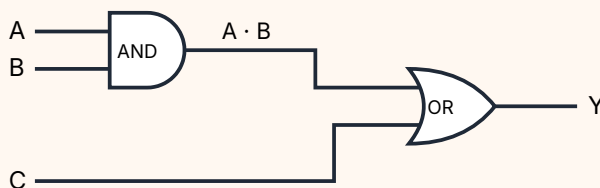
Remember: SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...



$Y = (A \cdot B) + C$: AND first, then OR.

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...

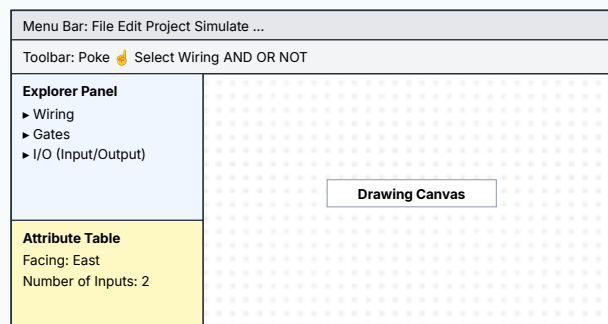
کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

LogiSim Evolution: Simulation Tool

LogiSim Evolution is a free graphical tool for designing and simulating logic circuits.

- "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change.
- It lets you see the truth tables and Boolean expressions you learn in theory.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.



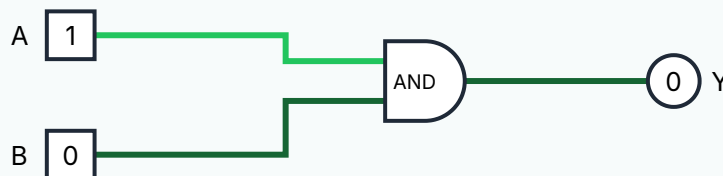
The main areas of the LogiSim Evolution window.

Remember: Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is..."

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔

Building a Circuit in LogiSim

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

Remember: Input pin = square (you push it). Output pin = circle (it shows a light). Poke = finger — you "poke" inputs to change them.

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائر...

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

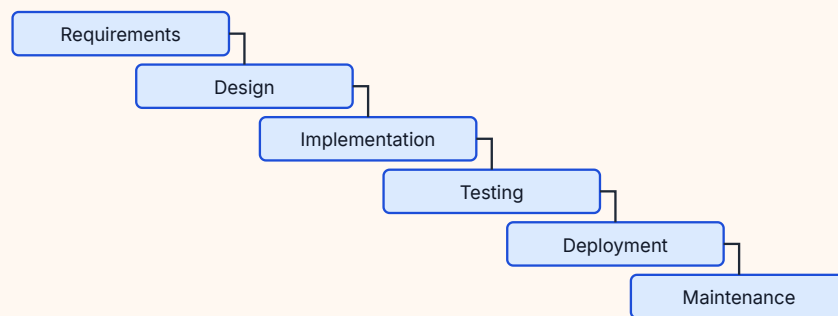
ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ...

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.



The Waterfall model: each phase "falls" into the next, in one direction only.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan→Design→Build→Test→Review & feedback→Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

Both models follow the same SDLC phases. The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases many times, in small loops.

- The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases...

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

Remember: One question decides it: "Will the requirements change?" No → Waterfall. Yes / not sure → Agile.

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

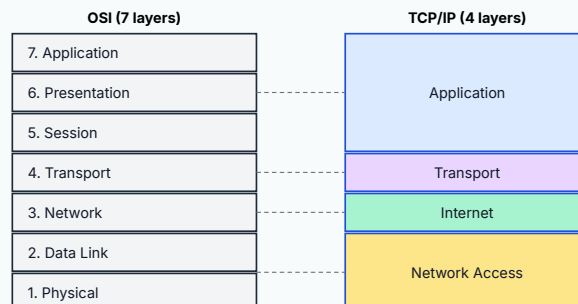
کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

The OSI model is mainly a reference model used for teaching and design.

- The TCP/IP model is the practical model used on the real Internet.
- It condenses the 7 OSI layers into just 4 layers.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Remember: Count it as 3-1-1-2: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total 3+1+1+2 = 7.

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام
Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالوگ سگنل (مسلسل اشارہ)
Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	ٹروٹھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل
Minterm	A product (AND) of all variables for a row with output 1.	منٹرم
Maxterm	A sum (OR) of all variables for a row with output 0.	میکس ٹرم
SOP	Sum of Products: minterms joined with OR.	حاصل ضرب کا مجموعہ
POS	Product of Sums: maxterms joined with AND.	مجموعوں کا حاصل ضرب
Karnaugh map (K-map)	A grid used to simplify Boolean expressions by grouping 1s.	کارنو نقشہ
Gray code	An ordering where neighbouring values differ by only one bit.	گرے کوڈ
Simulation	Running a computer model of a circuit to test it.	نقلی عمل (سمیولیشن)

Term	Meaning	اردو
SDLC	Software Development Life Cycle: the 6-phase process for building software.	سافٹ ویئر کی تیاری کا دورِ حیات
SRS	Software Requirements Specification: document listing what the software must do.	سافٹ ویئر ضروریات کی دستاویز
Black box testing	Testing inputs and outputs without seeing the code.	بلیک باکس جانچ
White box testing	Testing the internal logic while seeing the code.	وائٹ باکس جانچ
Waterfall model	A linear SDLC model; each phase is finished before the next.	آبشار ماڈل
Agile model	An iterative SDLC model using short sprints and constant feedback.	ایجنائل (لچکدار) ماڈل
Sprint	A short development cycle in Agile, usually 2–4 weeks.	اسپرینٹ (مختصر ترقیاتی دور)
Protocol	A set of rules for communication, e.g. HTTP, TCP.	پروٹوکول (ضوابطِ ابلاغ)
OSI model	ISO's 7-layer reference model of network communication (1984).	او ایس آئی ماڈل
TCP/IP model	The practical 4-layer model used on the Internet.	ٹی سی پی/آئی پی ماڈل

Chapter in one look

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (!) inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Computational Thinking & Algorithm

★ 6 Golden · 13 cards · Algorithm · Algorithm vs Pseudocode · Decomposition · Sorting Algorithms: Bubble Sort · Sorting Algorithms: Selection Sort · Searching Algorithms: Binary Search

2.1 Computational Thinking

Computational Thinking (CT) is a systematic approach to solving problems.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

- Decomposition — breaking a big problem down into smaller parts.
- Pattern recognition — identifying things that repeat or follow a rule.
- Abstraction — hiding (ignoring) details that do not matter.
- Algorithms — developing clear step-by-step instructions to solve the problem.

Remember: Remember the four parts of CT with the phrase "Don't Pick Any Apple": Decomposition, Pattern recognition, Abstraction, Algorithm.

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...).

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...).

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

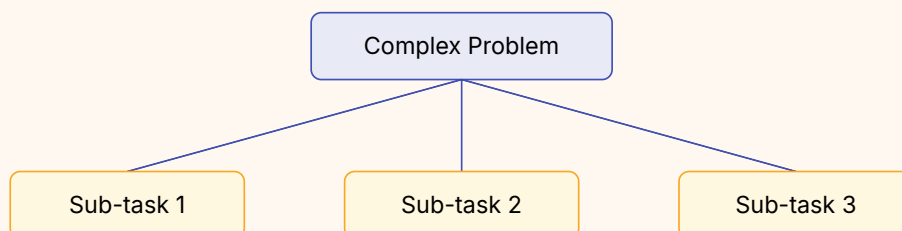
2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."



Decomposition: one complex problem is split into smaller sub-tasks.

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces. Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...).

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.3.2 Pattern Recognition

Pattern recognition is the process of identifying regularities, repetitions or rules in data.

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

Row	Stars
1	1
2	2
3	3
...	...
N	N

```
print("*")
print("**")
print("***")
print("****")
print("*****")
```

Remember: Whenever you see "the same thing again and again, with a small change each time", think loop. Pattern recognition is the habit of spotting that.

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔

2.3.3 Abstraction

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem.

Boil water → Add tea → Steep → Pour → Serve

- Necessary steps — the core logic that never changes.
- Changes based on the situation — these become variables (inputs) in the algorithm.
- Abstraction items (noise) — details you can ignore completely.

1. Necessary steps (core logic)	2. Changes based on situation (variables)	3. Abstraction items (ignore / noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few minutes 4. Pour into cup 5. Serve	Ingredients (black, green, milk) Additions (sugar, lemon) Tools (stove vs kettle)	Brand of kettle Cup colour or design Where tea is made (kitchen, office)

Remember: Abstraction = "Keep the skeleton, drop the decoration." Ask: "If I remove this detail, does the solution still work?" If yes, it is noise.

ایبسٹریکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپا کر صرف ضروری باتوں پر توجہ دینا۔

2.4 Sorting and Searching Algorithms

Sorting and searching are both fundamental techniques of computer science.

temp = A (temp = 8) → A = B (A = 4) → B = temp (B = 8)

- They work on lists of the same data type (for example, a list of all integers or a list of all names).
- Sorting/Searching Arranging a collection of items in an order.
- The order may be ascending (0–9, A–Z) or descending (9–0, Z–A).

Sorting	Searching
Arranging a collection of items in an order. The order may be ascending (0–9, A–Z) or descending (9–0, Z–A). Used to organise data.	Locating a specific element within a list to find its position or check its existence (is it there or not?).
Examples: Bubble Sort, Selection Sort	Examples: Linear Search, Binary Search
Real life: merit list arranged from highest to lowest marks.	Real life: finding your roll number in the merit list.

Trap: Position 1 in the textbook algorithm is index 0 in Python.

سارٹنگ (Sorting) کا مطلب ہے ڈیٹا کو صعودی (چھوٹے سے بڑے) یا نزولی (بڑے سے چھوٹے) ترتیب میں رکھنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position K → Compare DATA[K] with DATA[PTR], PTR = K+1 ... N → DATA[K] > DATA[PTR]? swap → End of pass: smallest is at K → K = K + 1

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K, the K-th smallest value sits in position K.

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	4 > 1 Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	1 > 9 No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	1 > 3 No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at outer + 1 and compares with the fixed position DATA[outer]. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

2.4.2 Searching Algorithms: Linear Search

Linear Search, also known as sequential search, works by examining each element in the list one by one, from the start, until the target is found (or the list ends).

PTR = first → DATA[PTR] = FindValue? → Yes: ValueFound = 1, exit → No: PTR = PTR + 1 → List ended? print "Not Found"

- It works on both sorted and unsorted data.
- A flag variable (here ValueFound) starts at 0 and becomes 1 when the item is found.
- After the loop, the flag tells us whether to print the item or "Not Found".

Step	Element checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not found
2	3	3 == 7 ?	Not found
3	6	6 == 7 ?	Not found
4	1	1 == 7 ?	Not found
5	7	7 == 7 ?	Found! (Exit loop)

Remember: Linear Search is like looking for your friend in a queue by checking every face from the front. Best case: 1 check (first person). Worst case: N checks (last...)

لینیئر سرچ (Linear Search) کو ترتیب وار تلاش (Sequential Search) بھی کہتے ہیں۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → MID = INT((BEG+END)/2) → Item = DATA[MID]? Found → Item < DATA[MID]: END = MID-1 → Else: BEG = MID+1 → Repeat while BEG ≤ END

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = INT((BEG + END) / 2) — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	INT(10/2) = 5	5	7 > 5	BEG = 6 (go right)
2	6	9	INT(15/2) = 7	7	7 = 7	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائنری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

2.5 Evaluation Criteria of Algorithm

Many algorithms can solve the same problem.

- To judge which is better, we evaluate them on three criteria: correctness, efficiency and clarity.

Input	Operation (divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient = 1, Remainder = 1)	No	Odd
6	6 / 2 (Quotient = 3, Remainder = 0)	Yes	Even
15	15 / 2 (Quotient = 7, Remainder = 1)	No	Odd

```
for number in [3, 6, 15]:
    remainder = number % 2
    if remainder == 0:
        print(number, "remainder", remainder, "→ Even")
    else:
        print(number, "remainder", remainder, "→ Odd")
```

Remember: Remember C-E-C: Correct (right answer, always finishes), Efficient (fast and light), Clear (easy to read).

کسی الگورتھم کو تین معیارات پر پرکھا جاتا ہے: درستگی (Correctness) یعنی درست نتیجہ دے، غیر معمولی صورتوں کو سنبھالے اور ہمیشہ ختم...

2.6 How to Decide Which Algorithm to Use

Before choosing an algorithm, ask three questions: What is the input size?

What is the task? → Need order? → Sorting algorithm → Need to find? Is data sorted? → No → Linear Search → Yes → Binary Search

- What is the input size? (10 items or 10 million?)
- Is the data sorted?
- Do we need speed or memory efficiency?
- If searching on unsorted data → use Linear Search.

Algorithm	Type	Best case	Worst case	Extra memory	Needs sorted data?	Stable?
Bubble Sort	Sorting	$O(N^2)$ as written; $O(N)$ with an early-stop check	$O(N^2)$	None — $O(1)$	—	Yes
Selection Sort	Sorting	$O(N^2)$	$O(N^2)$	None — $O(1)$	—	No
Linear Search	Searching	$O(1)$ (first item)	$O(N)$ (last item / absent)	None — $O(1)$	No	—
Binary Search	Searching	$O(1)$ (item at first MID)	$O(\log N)$	None — $O(1)$	Yes	—

Remember: Two keywords decide most exam answers: "sorted?" and "how big?". Unsorted or small → Linear. Sorted and big → Binary.

الگورتھم چننے سے پہلے دیکھیں کہ ڈیٹا کتنا بڑا ہے، ترتیب شدہ ہے یا نہیں، اور ہمیں رفتار چاہیے یا میموری کی بچت۔

Key terms

Term	Meaning	اردو
Computational thinking	A systematic approach to solving problems using decomposition, patterns, abstraction and algorithms.	کمپیوٹیشنل سوچ
Algorithm	A step-by-step logical procedure to solve a problem, independent of any language.	الگورتھم، مرحلہ وار طریقہ حل
Pseudocode	A human-readable, structured but non-executable representation of an algorithm.	سوڈو کوڈ، فرضی کوڈ
Decomposition	Breaking a complex problem into smaller, manageable parts.	تقسیم کاری
Sub-task	One small part of a decomposed problem.	ذیلی کام
Pattern recognition	Identifying regularities, repetitions or rules in data.	نمونے کی پہچان
Abstraction	Hiding unnecessary details and focusing on essentials.	تجربہ، غیر ضروری تفصیل چھپانا
Noise	Details that do not affect the solution and can be ignored.	فالتو تفصیلات
Sorting	Arranging items in ascending or descending order.	ترتیب دینا
Searching	Locating an element in a list to find its position or existence.	تلاش کرنا
Ascending order	Smallest to largest (0–9, A–Z).	صعودی ترتیب
Descending order	Largest to smallest (9–0, Z–A).	نزولی ترتیب
Bubble Sort	Sort that repeatedly compares and swaps adjacent elements.	بل سارٹ
Selection Sort	Sort that repeatedly selects the smallest element and puts it at the front.	سلیکشن سارٹ
Pass	One full run of the inner loop through the list.	ایک چکر
Swap	Exchanging the values of two positions, usually with a temp variable.	جگہ کا تبادلہ
In-place sorting	Sorting without needing extra memory for another list.	اضافی میموری کے بغیر ترتیب

Term	Meaning	اردو
Stable sort	A sort that keeps equal items in their original order.	مستحکم ترتیب
Linear (sequential) search	Checking each element one by one until the target is found.	ترتیب وار تلاش
Binary search	Search that halves a sorted list at each step.	دو حصی تلاش
BEG / MID / END	First, middle and last positions of the part being searched in Binary Search.	آغاز / درمیان / اختتام
Flag	A variable (like ValueFound) that records whether something has happened.	نشان دہی کرنے والا متغیر
Correctness	Giving expected output, handling edge cases and always finishing.	درستگی
Efficiency	How well an algorithm uses time and memory as input grows.	کارکردگی
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Trace table	A table recording values step by step during a dry run.	ٹریس ٹیبل، جانچ جدول
Edge case	An unusual or extreme input used to test an algorithm.	انتہائی صورت
Time complexity (Big O)	A measure of how the number of steps grows with input size, e.g. $O(N)$, $O(\log N)$.	وقتی پیچیدگی

Chapter in one look

- **Computational thinking** is systematic problem solving using decomposition, pattern recognition, abstraction and algorithms.
- An **algorithm** is a step-by-step logical procedure to solve a problem; it is conceptual and language-independent.
- Algorithms give structure, optimise time and memory, are reusable, ensure accuracy and act as a blueprint for code.
- **Pseudocode** is a human-readable, structured representation of an algorithm using keywords like IF, FOR and WHILE; it is not executable.
- An algorithm defines *what* to do; pseudocode shows *how* to implement it.
- **Decomposition** breaks a big problem into smaller sub-tasks; it reduces complexity, helps debugging and promotes reusability.
- **Pattern recognition** finds repetitions and rules (Row $N = N$ stars) and is the foundation of loops.
- **Abstraction** hides noise and keeps only essential steps; situational details become variables.
- Sorting arranges data (ascending or descending); searching locates an element's position or existence.
- **Bubble Sort** compares and swaps adjacent elements; the largest value moves to the end each pass; $N - 1$ passes; stable; $O(N^2)$.
- **Selection Sort** brings the smallest remaining value to the front each pass; inner loop starts at outer + 1; fewer swaps; not stable; $O(N^2)$.
- **Linear Search** checks items one by one; works on unsorted data; $O(N)$ worst case.
- **Binary Search** halves the search range using BEG, MID and END; needs sorted data; $O(\log N)$ — about 20 steps for 1 million items.
- Algorithms are evaluated on **correctness** (checked with trace tables and edge cases), **efficiency** (time and space) and **clarity**.
- Choose by asking: input size, sorted or not, speed or memory? Unsorted → Linear; sorted → Binary; needs order → sort.

Programming Fundamentals

★ 12 Golden · 24 cards · Fundamentals of Python Programming · Variables · Data Types in Python · Operators and Operands: Arithmetic and Assignment · Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Selection Statements: if and if-else · Selection Statements: if-elif-else and Nested if · Repetition Statements: the for Loop and range() · The while Loop, break and continue · Nested Loops and for vs while

3.1 Programming

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own.

- A set of instructions given to a computer is called a program.
- The language used to write these instructions is called a programming language.
- Writing programs is called programming, and the person who writes them is a programmer.

```
print("Hello")
```

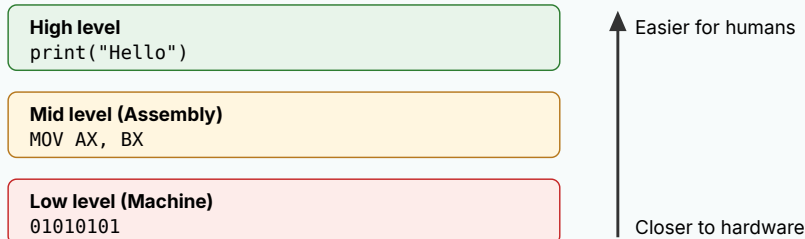
Remember: Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix...

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

3.1.1 Programming Language Types

Programming languages are classified into three main types, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. MOV, ADD).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Remember: "Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الف...

3.2 Career Growth with Python

Python is one of the most in-demand languages in the world.

- Software Development: build desktop and web applications.
- Data Science and Data Analysis: analyse data, create reports and make predictions.
- Artificial Intelligence and Machine Learning: develop intelligent systems, chatbots and recommendation algorithms (like "videos you may...")
- Web Development: use frameworks such as Django and Flask to build websites.

```
# Python
print("Hello")
```

Remember: Six careers — remember "S-D-A-W-A-C": Software, Data, AI, Web, Automation, Cybersecurity.

پائتھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے ہیں۔

3.3 Integrated Development Environment (IDE)

An IDE (Integrated Development Environment) is a software tool that helps programmers write, run and manage code in one place.

- Code Editor: where you write and edit code.
- Syntax Highlighting: colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- Auto-completion: suggests code while you type (type `print` and it offers `print`).
- Run/Execute Tool: runs the program with one click.

Remember: IDE features — "E-S-A-R-D": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

3.3.1 Visual Studio Code (VS Code)

VS Code is a free, lightweight and powerful code editor made by Microsoft and widely used for Python.

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) → Explorer: new file `hello.py` → Type code → Click ► Run

- By default it is not a full IDE, but it becomes one when you install extensions (add-ons), such as the Python extension by Microsoft.

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

```
print("Hello, World!")
```

Remember: Extensions are the "App Store" of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	int	Whole numbers without decimals (positive, negative or zero).	age = 16
Floating point	float	Numbers with a decimal point.	height = 5.7
String	str	Text data, enclosed in quotes (" " or ' ').	name = "Sammy" or year = "2026"
Boolean	bool	Logical value: True or False (capital T / F).	is_student = True
List	list	Ordered, changeable collection in square brackets [].	marks = [85, 90, 75]
Tuple	tuple	Ordered, unchangeable sequence in parentheses ().	coords = (10, 20)
Dictionary	dict	Key : value pairs in curly braces { }.	student = {"name": "Ahmed", "age": 16}
Set	set	Unordered collection of unique items (no duplicates).	vowels = {'a', 'e', 'i'}

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.2 Input and Output Handling

Output — the print() function. A function is a named, ready-made piece of code that does a job when you call it with brackets.

- Items separated by commas are printed with one space between them.
- An f-string (a string with f before the quote) lets you put variables directly inside { }. It is the fastest and most readable way to...

Remember: "input() always gives a string." If you plan to do maths, wrap it: int(input(...)) for whole numbers, float(input(...)) for decimals. Read it from the inside...

اسکرین پر معلومات دکھاتا ہے اور input() صارف سے معلومات لیتا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In $10 + 20$: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	$5 + 3$	8
-	Subtraction	$10 - 4$	6
*	Multiplication	$4 * 2$	8
/	Division (result is always a float)	$9 / 2$	4.5
//	Floor division (drops the decimal part)	$9 // 2$	4
%	Modulus (remainder)	$9 \% 2$	1
**	Exponentiation (power)	$2 ** 3$	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when $n \% 2 == 0$.

Exam: You will be asked to evaluate expressions such as $17 // 5$, $17 \% 5$, $2 ** 4$ or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپریٹنڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	$a = b$	False
≠	Not equal to	$a \neq b$	True
>	Greater than	$a > b$	True
<	Less than	$a < b$	False
≥	Greater than or equal to	$a \geq 10$	True
≤	Less than or equal to	$a \leq b$	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate $a > b$ and $b > 3$ for $a = 10$, $b = 5$ " or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: == compares,...

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

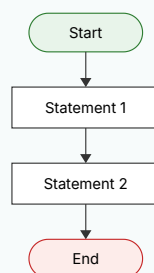
Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.1 Sequence

Sequence means statements are executed one after another, strictly from top to bottom, in the exact order they are written.

- This is the default behaviour of Python.
- No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
```

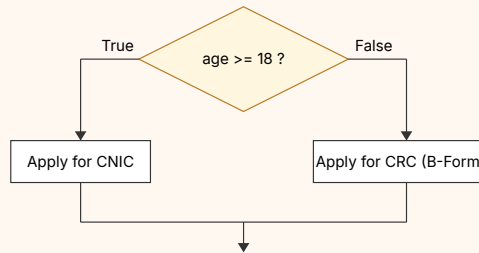
Remember: Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern Input → Process → Output.

سیکوننس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.



An if-else statement: one diamond (condition), two paths, then the paths join again.

```
if condition:
    statement(s)    # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input()),...`

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions. (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):
    print(i)
```

Remember: range stops one short: `range(1, 11)` gives 1 to 10. To include a number `n`, write `range(..., n + 1)`. Number of values in `range(a, b)` = `b - a`.

Exam: Expect "predict the output" questions on `range()` with a step, and programs such as "print a table", "sum of first `n` numbers", "even numbers from 1 to 20". Always state...

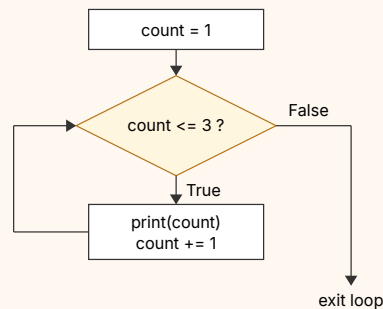
لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends



A while loop: test the condition, run the body, go back and test again.

```

count = 1
while count <= 3:
    print(count)
    count += 1  # VERY IMPORTANT: update the condition!
  
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i <= 3: / print(i) / i += 1</code>

```

for outer in range(1, 3):          # runs 2 times (1, 2)
    for inner in range(1, 4):      # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
  
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

Practical Activity: Shopping Bill Calculator

This program combines all three control structures: sequence (steps in order), selection (checking "done", checking the price, checking for a discount) and repetition (a while loop that keeps adding items).

- It uses a flag — a Boolean variable (shopping) that controls when the loop stops.
- item.lower() converts the text to lowercase so that "done", "Done" and "DONE" all work.

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

Remember: Label the parts of any program with comments, as here: # REPETITION, # SELECTION. It helps you, your teacher and the examiner see the structure.

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، if-else سے غلط قیمت رو...

3.6 Libraries in Python

A library (also called a module) is a collection of pre-written code that offers ready-made functions.

- Instead of writing a square-root algorithm from scratch, you import math and call math.sqrt().
- Libraries save time, reduce bugs and let you stand on other programmers' work.
- There are five common ways to import a library.

Method	Syntax	How you call the function
1. Entire module	import math	math.sqrt(25)
2. Alias (nickname)	import math as m	m.sqrt(16)
3. Specific function	from math import sqrt	sqrt(36)
4. Several functions	from math import sqrt, pi	sqrt(9), pi
5. Everything (*)	from math import *	sqrt(64) — avoid this

Trap: from math import * dumps every name into your program.

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے import سے پروگرام میں لایا جاتا ہے۔

3.6.1 Built-in Libraries

Built-in libraries come pre-installed with Python.

- You only import them; you do not run pip.
- Three that every Class XI student must know are math, random and datetime.

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

Remember: ceil always rounds up (ceiling); floor always rounds down. round() (built-in, not in math) goes to the nearest integer and is a different function.

math حساب کے لیے، random بے ترتیب عدد کے لیے اور datetime تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔

3.6.2 Third-Party Libraries

Third-party libraries are written by the community.

- They are not bundled with Python, so you install them from a terminal with pip install library_name and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

```
pip install pandas
import pandas as pd
```

Trap: pip is a terminal command, not a Python statement.

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں pip install سے انسٹال کیا جاتا ہے۔

3.7 Debugging Code in Python

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly.

- Every program you write will have bugs.
- The skill is not "never make a mistake"; it is "find the mistake quickly".
- Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>
Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

```
x = int("abc")
```

Remember: Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

Exam: Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong...

ڈیبنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔

Key terms

Term	Meaning	اردو
Program	A set of instructions a computer can follow.	پروگرام
High-level language	A language close to English (Python, Java, C++).	اعلیٰ سطحی زبان
Interpreter	Runs a program line by line without a separate compile step.	انٹرپریٹر
IDE	Software that combines editor, terminal and debugger.	انٹیگریٹڈ ڈویلپمنٹ انوائرنمنٹ
Variable	A named box that holds a value.	متغیر
Data type	The kind of value: int, float, str, bool, list, dict.	ڈیٹا کی قسم
Operator	A symbol that performs an operation on operands.	عامل
Operand	A value an operator works on.	معمول
Sequence	Instructions executed from top to bottom.	تسلسل
Selection	Choosing a path with <code>if / else</code> .	انتخاب
Repetition / loop	Repeating a block with <code>for</code> or <code>while</code> .	تکرار / لوپ
Nested	A structure placed inside another of the same kind.	تو در تو
Library / module	A collection of ready-made functions.	لائبریری / ماڈیول
pip	The tool that installs third-party Python libraries.	پپ
Syntax error	A grammar mistake; the program will not start.	نحوی غلطی
Runtime error	A crash while the program is running (exception).	رن ٹائم غلطی
Logical error	The program runs but the result is wrong.	منطقی غلطی
Traceback	Python's report of where a crash happened.	ٹریس بیک
Flag	A Boolean variable that controls a loop or decision.	فلیگ
Integer division	<code>//</code> division that drops the fraction.	صحیح تقسیم

Chapter in one look

- A program is a set of instructions; programming languages are low-level (close to hardware) or high-level (close to English). Python is a high-level, interpreted, readable language.
- Python is widely used in web development, data science, AI and automation — skills that grow a career in Pakistan and abroad.
- An IDE (such as VS Code plus the Python extension) combines an editor, a terminal, a debugger and syntax highlighting.
- Variables store values; Python types are dynamic. Core types: `int`, `float`, `str`, `bool`, `list`, `dict`.
- `print()` displays output; `input()` always returns a **string** — convert with `int()` or `float()` before arithmetic.
- Operators: arithmetic (`+` `-` `*` `/` `//` `%` `**`), assignment (`=` `+=` `...`), relational, logical (`and` `or` `not`), membership (`in`), bitwise (`&` `|` `^` `~` `<<` `>>`).
- Control structures: sequence (top to bottom), selection (`if` / `if-else` / `if-elif-else` / nested), repetition (`for` / `while`).
- `range(stop)` goes 0 ... stop-1; `range(start, stop, step)` jumps by step. `break` leaves a loop; `continue` skips one iteration.
- Use `for` when the count is known; use `while` when you wait for a condition (and always update that condition).
- Libraries extend Python. Built-in: `math`, `random`, `datetime`. Third-party (need `pip`): NumPy, Pandas, Matplotlib, TensorFlow.
- Errors: syntax (grammar, will not run), runtime (crash, traceback), logical (runs, wrong answer).

CHAPTER 4

Data and Analysis

★ 4 Golden · 15 cards · Keys and Integrity Constraints · Entity Relationship Model (ER-Model) · Referential Integrity · Creating Queries in MS Access

4.1.1 Data, Information and Database

Data are unprocessed facts, figures or observations.

Data → Processing → Information → Decision

- They can be quantitative (numbers: 80, 75, 90) or qualitative (descriptions: the colours of cars in a parking lot).
- Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.
- When you organise, process and interpret data, it becomes information.

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

Remember: Exam line: Data is raw; information is processed data used for decision-making. Always add one pair of examples.

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔

4.1.2 Database Management System (DBMS)

A DBMS is software that lets you create, manage and interact with databases.

- Less redundancy. The department name is stored once, not in every student row.
- Consistency. Change a phone number once; every screen that uses it shows the new number.
- Security. Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- Integrity. Rules stop duplicate roll numbers and empty primary keys.

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

4.2 Database Components (Tables, Records, Fields)

The basic structure is a table (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

- A record (row, tuple) is one complete instance — everything about Ali.
- A field (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

Remember: Memory: row = record = one person. column = field = one question on the admission form.

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

کنڈیڈٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.3 Relational Database Management System (RDBMS)

An RDBMS stores data in several tables and uses primary and foreign keys to keep relationships.

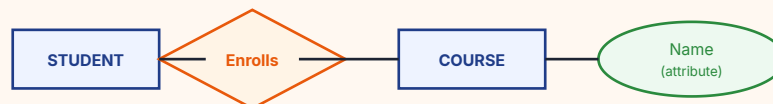
- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with SQL (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیڑ سے ان کے تعلقات برقرار رکھتا ہے۔

4.4 Entity Relationship Model (ER-Model) ★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.6 Relational Schema Development

A relational schema is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate.

- You write it before clicking around in Access.
- It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships.
- Benefits: less redundancy, integrity planned in advance, easier to grow later.

```
Department (DeptID PK, DeptName)
Student (RollNo PK, Name, Class, DeptID FK → Department)
Course (CourseID PK, Title, Credit, DeptID FK → Department)
```

Remember: Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

4.7 Case Study — Library Management System

A college library needs a database. Walk the seven steps examiners expect.

- A college library needs a database.
- Walk the seven steps examiners expect.
- A book has ID, title, author, publisher.
- A member (student/teacher) has ID, name, contact.

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	



Final library ER: two 1:M relationships through the BorrowRecord junction table.

```
Member (MemberID PK, Name, ContactInfo)
Book (BookID PK, Title, Author, Publisher)
BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)
```

لائبریری میں ممبر اور کتاب کا تعلق بہت-بہت-سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی فارن کیز اور اجراء...

4.8 Database Objects (Tables, Forms, Queries, Reports)

Four objects you will meet in Access. Learn the one-line jobs cold:

- Learn the one-line jobs cold:

Object	Job	You ...
Table	Stores the data (foundation; every other object depends on it)	keep the actual records here
Form	Friendly screen for enter / edit / view (text boxes, drop-downs)	type data in; fewer mistakes than a raw grid
Query	Ask a question: filter, sort, calculate	e.g. "students above 80%"
Report	Formatted output for printing or sharing	read only — never type into a report

Remember: Store → Enter → Search → Print. Tables, Forms, Queries, Reports.

ٹیبل ڈیٹا رکھتی ہے، فارم سے اندراج ہوتا ہے، کوئری سوال پوچھ کر چھانتی ہے، رپورٹ چھپائی کے لیے سنواری ہوئی پیشکش ہے — اس میں ڈیٹا بھرا نہیں جاتا۔

4.9 Creation of Tables in Microsoft Access

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

- Create tab → Table.
- A blank grid appears.
- Double-click Field1, Field2, ... to rename.
- Type data; Access guesses the data type.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

Trap: Phone numbers look numeric but are Short Text.

پہلے ٹیبل کا منصوبہ بنائیں۔

4.10 Designing Forms for Data Manipulation

A form is the screen people actually type into.

- Create tab → Form Wizard.
- Select the table (e.g. Student).
- Move fields with > (one) or >> (all).
- Choose layout. Columnar = one record at a time, best for data entry.

فارم اندراج کی سادہ اسکرین ہے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

4.11.4 Data Summarization and Statistical Analysis

Grouping combines rows that share a value (all CS students together) and then an aggregate calculates a number for the group.

- In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT، SUM، AVG، MAX، MIN سے خلاصہ نکلتا ہے۔

4.11.6 Data Visualization in MS Access (Charts)

Charts turn numbers into a picture so a pattern jumps out.

- Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)



Column = compare categories. Pie = share of a total. Line = change over time.

Exam: "Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔

Key terms

Term	Meaning	اردو
Data	Raw facts and figures.	ڈیٹا / خام حقائق
Information	Processed data that supports a decision.	معلومات
Database	Organised collection of related data.	ڈیٹا بیس
DBMS	Software that creates and manages databases.	ڈیٹا بیس مینجمنٹ سسٹم
Table	Rows and columns about one entity.	ٹیبل
Record	One row — one instance.	ریکارڈ
Field	One column — one attribute.	فیلڈ
Primary key	Chosen unique, non-empty identifier.	پرائمری کی
Foreign key	Field that refers to another table's PK.	فارن کی
Candidate key	Any field(s) that could be the PK.	کینڈیڈیٹ کی
Alternate key	A candidate not chosen as PK.	آلٹرنیٹ کی
RDBMS	DBMS that stores related tables linked by keys.	ریلیشنل ڈی بی ایم ایس
SQL	Language to query relational databases.	ساختہ استفساری زبان
Entity	Object we store data about (ER rectangle).	ہستی
Attribute	Property of an entity (ER oval).	صفت
Relationship	Link between entities (ER diamond).	تعلق
Cardinality	1:1, 1:M or M:N.	عددییت
Junction table	Bridge table that splits an M:N into two 1:M.	جنکشن ٹیبل
Referential integrity	Every FK matches a live PK (no orphans).	حوالہ جاتی درستگی
Cascade Update	Parent PK change copied to child FKs.	کیسکیڈ اپ ڈیٹ
Cascade Delete	Deleting the parent deletes related children.	کیسکیڈ ڈیلیٹ
Schema	Written blueprint of tables, keys, relationships.	اسکیم / نقشہ
Form	Screen for entering data.	فارم
Query	Saved question that filters/calculates data.	کوئری
Report	Formatted printable output.	رپورٹ
AutoNumber	Access-generated unique integer, often the PK.	آٹو نمبر

Chapter in one look

- Data = raw facts; information = processed data used for decisions; a database stores related data in a structured, updatable way.
- A DBMS is the software layer that provides less redundancy, consistency, security and integrity.
- Table = rows (records) and columns (fields). One table, one entity.
- Candidate keys could identify a row; the primary key is the one you choose (unique, not NULL); leftover candidates are alternate keys; a foreign key points at another table's PK.
- An RDBMS keeps several related tables and is queried with SQL.
- ER symbols: rectangle entity, oval attribute, diamond relationship. Cardinality 1:1, 1:M, M:N. M:N needs a junction table.
- Referential integrity forbids orphan FKs. Cascade Update copies a changed PK into children; Cascade Delete removes children with the parent.
- Write a relational schema before building. Library case: Member and Book are M:N, resolved by BorrowRecord.
- Objects: tables store, forms enter, queries search, reports print.
- Access Design View sets field names, data types and the PK. Phone numbers are Short Text; money is Currency; auto IDs are AutoNumber.
- Forms need a table/query source; Wizard + columnar layout for data entry.
- Query criteria: one value; AND on the same row; OR on two rows. Totals Σ for Group By / Count / Sum / Avg / Max / Min.
- Bar/column compare categories, pie shows parts of a whole, line shows a trend over time.

CHAPTER 5

Application and Impacts of Computing

★ 4 Golden · 13 cards · Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Information Sources (Primary, Secondary, Tertiary) · Assistive Technologies

5.1 Introduction to Computing Impacts

Computing means using computers and computer-based devices to store, process and share information.

- Schools and colleges — online admissions, e-books, online results of the Sindh boards.
- Hospitals — computerised patient records, digital X-rays, online appointment systems.
- Banks — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- Farms — weather forecasts on mobile phones, soil sensors, tube-well control.

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

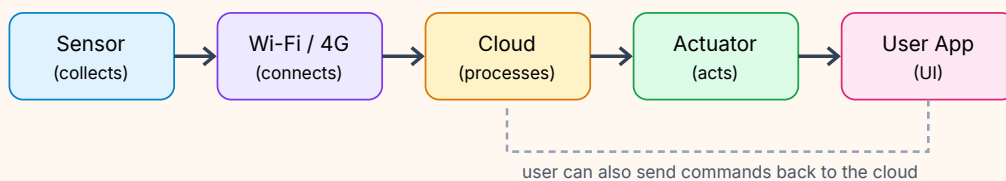
5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information. Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands



The six IoT components working as one system (data processing happens inside the cloud).

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

5.1.3 Data Analytics & Components

Data Analytics is the process of collecting, organising, cleaning, studying and presenting data to find useful information.

1. Collection → 2. Storage → 3. Cleaning → 4. Analysis → 5. Visualization

- It helps people make better decisions.
- Data means raw facts and figures (for example, "72, 45, 88").
- Information is data that has been processed so it has meaning ("Ali scored 72%; he passed").

Roll No	Name	Marks (raw)
1	Ayesha	42
2	Bilal	18
3	Hina	35
3	Hina	35
4	Kashif	250
5	Sana	(missing)

Remember: "Cows Sit Calmly And Vote" — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar...

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔

Comparison: IoT vs AI vs Data Analytics

These three technologies are often used together, so students mix them up.

- The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

Remember: Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

5.2 Uses of IoT in Different Areas

IoT improves efficiency (doing more with less effort), saves time, and enhances the overall quality of life.

Soil Sensor → Cloud AI (Needs water?) → Water Pump

- It plays a significant role in developing modern infrastructure (the basic systems of a country — roads, power, water, hospitals).
- The source lists six areas.

Remember: Remember the six areas with "CHIEF H": Cities, Health, Industry, Education, Farming, Homes.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔

5.2.1 Scope of AI in Education in Pakistan

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving teaching, learning and administration (managing the school: admissions, records, timetables).

- Learning Management Systems (LMS) — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI...
- Automated grading and digital testing — MCQ tests marked instantly by computer.
- AI-based career counselling — software that studies a student's marks and interests and suggests suitable fields.
- Helps students learn at their own speed.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

Trap: Mistake: Writing that AI will replace teachers.

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔

5.2.2 Career Connection in IoT and AI

Learning computing skills opens doors to many employers in Pakistan: software houses (e.g.

- IoT Engineer: designs smart systems and connected devices (e.g. a smart irrigation kit).
- Data Analyst: analyses the data generated by IoT systems and produces reports.
- Programming — Python, C++.
- Networking — Wi-Fi, protocols (rules for how devices communicate).

Remember: Skills = "PNEP": Programming, Networking, Electronics, Problem-solving. You start building the first one this year when you learn Python!

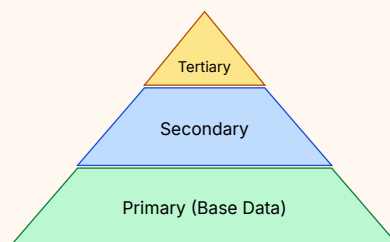
کمپیوٹنگ کی مہارتیں سیکھ کر آپ سافٹ ویئر ہاؤسز، ٹیلی کام کمپنیوں (جیسے جاز، ٹیلی نار)، بینکوں اور سرکاری آئی ٹی محکموں میں کام کر سکتے ہیں۔

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short),...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

5.4 Impacts of Computing in Various Fields

Computing has drastically changed society.

- It makes life faster and more connected, but it also brings challenges like privacy risks and technology addiction.
- For every field, think of both sides.

Field	Positive impact (+)	Negative impact (-)
Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Remember: For the five fields, remember "E-C-B-H-A" — Education, Communication, Business, Healthcare, Agriculture — and always write one + and one – for each.

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___".

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور مع...

5.5.1 Importance of Assistive Technologies

Assistive technologies are important for three main reasons: Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).

Equal Access → Independence → Inclusion

- Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
- Independence in daily life: Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the...
- Social inclusion: They break down barriers, allowing individuals to work, study and socialise confidently.

Remember: Remember "All": Access, Independence, Inclusion.

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی می...

5.5.2 Career Connection in Assistive Tech

Designing assistive technology is a highly rewarding career path — you earn a living while improving people's lives.

- CareerWhat they do Assistive Software DeveloperCreates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
- Biomedical EngineerDesigns artificial limbs and hearing aids.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔

Key terms

Term	Meaning	اردو
Computing	Using computers and digital devices to store, process and share information.	کمپیوٹنگ
Artificial Intelligence (AI)	Human-made technology that lets machines think, learn and decide using data.	مصنوعی ذہانت
Internet of Things (IoT)	Physical devices connected to the internet that collect and share data to work automatically.	انٹرنیٹ آف تھنگز، اشیاء کا انٹرنیٹ
Sensor	Device that collects information from the physical world.	حساس آلہ (سینسر)
Actuator	Device that performs a physical action in response to a command.	عمل کار آلہ (ایکچویٹر)
Connectivity	The link (Wi-Fi, Bluetooth, ZigBee, 4G/5G) that lets devices send and receive data.	رابطہ کاری
Data processing	Changing raw data into useful information.	ڈیٹا کی پروسیسنگ
Cloud computing	Storing data and using services over the internet instead of local drives.	کلاؤڈ کمپیوٹنگ
User interface (UI)	The part that lets a user see information and give commands to a device.	صارف انٹرفیس
Data Analytics	Collecting, storing, cleaning, analysing and presenting data to find useful information.	ڈیٹا کا تجزیہ
Data cleaning	Finding and correcting errors such as duplicates and missing values.	ڈیٹا کی صفائی
Data visualization	Showing data as charts, plots or graphs.	ڈیٹا کی بصری پیشکش
Smart city	City using IoT for traffic, lighting, parking and waste.	اسمارٹ شہر
Smart farming	Using sensors, automated irrigation and drones in agriculture.	اسمارٹ کاشتکاری
Learning Management System (LMS)	Online platform for lessons, assignments and tests.	آن لائن تعلیمی نظام
Telemedicine	Treating patients remotely using computing and communication.	دور دراز علاج
Primary source	Original, firsthand information.	بنیادی ذریعہ

Term	Meaning	اردو
Secondary source	Information collected and explained by someone else.	ثانوی ذریعہ
Tertiary source	Highly summarised information for quick reference.	ثالثی ذریعہ
Reliability	How far a source can be trusted.	قابل اعتماد ہونا
Privacy	Keeping personal information safe from others.	رازداری
Cyber-attack	An attempt by hackers to damage or steal from computer systems.	سائبر حملہ
Assistive technology	Devices, software and systems that help people with disabilities.	معاون ٹیکنالوجی
Screen reader	Software that reads screen text aloud for blind users.	اسکرین ریڈر
Braille display	Device with raised pins showing text as Braille dots.	بریل ڈسپلے
Accessibility	Designing so everyone, including people with disabilities, can use a product.	رسائی پذیری

Chapter in one look

- Computing is vital in Pakistan's schools, hospitals, banks, farms and industries; it has both positive and negative impacts.
- **AI** is "artificial" (made by humans) and "intelligent" (does tasks needing thinking); it uses data to learn, recognise patterns and make decisions.
- In smart systems, IoT is the "eyes" that gather data, Analytics sorts it, and AI is the "brain" that decides the action.
- **IoT** is a system of physical devices connected to the internet that collect and share data to perform tasks automatically.
- The six IoT components are sensors (collect), actuators (act), connectivity, data processing, cloud computing and user interface.
- **Data Analytics** is a pipeline: collection → storage → cleaning → analysis → visualization.
- Data cleaning removes duplicates, incorrect values, missing records and improper formats.
- IoT outputs raw data (hardware); AI outputs smart decisions (software); Data Analytics outputs reports, graphs and insights (data).
- IoT is used in smart cities, industrial automation, healthcare, education, home automation and smart farming.
- Smart farming (soil sensor → cloud AI → water pump) saves water and improves yields in Sindh and Punjab.
- AI in Pakistani education: LMS, automated grading, career counselling; challenges are rural internet, lack of devices, teacher training and data privacy.
- IoT/AI careers (IoT engineer, data analyst) need programming, networking, electronics and problem-solving.
- Information sources: primary (firsthand/original), secondary (collected and explained by others), tertiary (summarised for quick reference).
- Do not trust an online article that lacks an author, a publication date or references.
- Computing helps education, communication, business, healthcare and agriculture but risks screen addiction, isolation, job loss, cyber-attacks and privacy loss.
- **Assistive technologies** — mobility, sensory, communication and cognitive aids — give equal access, independence and social inclusion to people with disabilities.

Digital Literacy

★ 11 Golden · 17 cards · Types of Data · Data-Collection Strategies · Interviews and Surveys · Prototypes, Observation and Simulations · Primary and Secondary Data · Case Study: Digital Inquiry Project · Steps 1 – 3: Advanced Search and Methodology · Steps 4 – 6: Survey Preparation and Primary Data Collection · Steps 7 – 9: Secondary Data and Spreadsheet Organisation · Steps 10 – 12: Data Analysis, Conclusion and Recommendations · The Final Digital Artefact

6.1 Introduction to Digital Literacy

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a safe, effective, and responsible manner.

Locate information → Organise it properly → Turn it into something useful → Share it responsibly

- Safe – you protect yourself and others online (for example, you do not share passwords or private details).
- Effective – you get the job done well (for example, you find the right information quickly).
- Responsible – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).
- find information,

Remember: Remember the four verbs of a digitally literate student: Locate – Organise – Create – Share ("LOCS": think of a lock that keeps your digital life safe).

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔

6.2 Digital Literacy in the Information Age

The Information Age is the period we live in now, where vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms (websites, apps, social media).

- Simply having access to technology is not enough.
- Owning a smartphone does not make you good at using information.
- You must develop the skills to understand, manage and use information wisely.

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Remember: Three abilities = "Tech, Think, Talk": Technical, Cognitive (think), Social (talk with others).

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل...

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews-Surveys-Prototypes-Observation-Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? —	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جاسکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Primary Data

Fresh, direct collection
(You do the work)

Surveys · Interviews · Observation

Secondary Data

Existing archives
(Someone else did it)

Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.6 Designing a Data-Collection Approach

Before collecting data, a researcher must have a plan.

Question → Data type → Method → Sample → Tools → Collect → Store + backup → Ethics

- Define the research question clearly. What exactly do you want to find out?
- Decide the type of data needed: quantitative, qualitative, or both.
- Select suitable methods (survey, interview, observation).
- Choose the sample or target group – the people who will give you data.

Remember: Remember the steps with: "Quiet Teachers Make Students Think, Check, Save Ethically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں...

6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

Collecting data is only one part of the work.

Spreadsheet data → Chart / Graph → Insert → Slides

- You must also present your findings so that others can understand them quickly and easily.
- Digital tools make this much easier.
- The four main tools are spreadsheets, presentations, infographics and reports.

Formula	What it does
=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4, "Yes")	Counts cells that match a condition.

Remember: Presentation rule of thumb: one idea per slide, and no more than about six short bullet points. If you are reading long sentences off the slide, the slide has...

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔

6.7.3 – 6.7.4 Infographics and Reports

An infographic (information + graphic) is a visual way of presenting information, facts or data so that people can understand the main idea quickly and easily.

- Simplify complex concepts.
- Highlight the most important findings and results.
- Use clear visuals, limited text and accurate information.

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

Remember: Match tool to job: Spreadsheet = Calculate, Presentation = Tell, Infographic = Show quickly, Report = Explain fully.

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva) اور پاور بی آئی...

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1 – 3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students" " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (" ") بالکل وہی جملہ تلا...

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most? Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

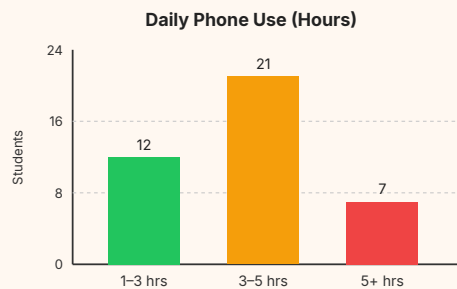
چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابل اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے مواز...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option Suitable digital tools Main purpose Digital Report MS Word, Google Docs Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

Careers in Digital Research

The skills in this chapter lead to real jobs: CareerWhat they doChapter skill used Market Research AnalystDesigns surveys to collect primary data to guide business decisions.Surveys, primary data UX (User Experience)...

Career	What they do	Chapter skill used
Market Research Analyst	Designs surveys to collect primary data to guide business decisions.	Surveys, primary data
UX (User Experience) Researcher	Gathers qualitative and quantitative data to improve app and website designs.	Prototypes, interviews, observation
Data Journalist	Uses advanced search skills to research stories and presents them via infographics.	Advanced search, infographics
Digital Marketing Specialist	Collects audience data and presents results through dashboards and reports.	Dashboards, reports

Remember: Link each career to one tool: Market Researcher → Survey; UX Researcher → Prototype; Data Journalist → Search + Infographic; Digital Marketer → Dashboard.

اس باب کی مہارتوں سے کئی کیریئر بنتے ہیں: مارکیٹ ریسرچ اینالسٹ کاروباری فیصلوں کے لیے سروے بناتا ہے، یو ایکس (UX) ریسرچر ایپس اور...

Key terms

Term	Meaning	اردو
Digital literacy	Ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly.	ڈیجیٹل خواندگی
Information Age	Period in which vast amounts of data are produced, shared and accessed instantly.	معلوماتی دور
Data	Facts, numbers, words or observations collected for study.	مواد / ڈیٹا
Qualitative data	Descriptive, non-numerical data about thoughts, feelings and experiences.	کیفیتی ڈیٹا
Quantitative data	Numerical data that can be counted or measured.	مقداری ڈیٹا
Interview	Planned conversation to gather information directly from a person or small group.	انٹرویو / بالمشافہ گفتگو
Survey / Questionnaire	Same set of questions asked to many people.	سروے / سوالنامہ
Prototype	Simple early version of a product used to test an idea and get feedback.	ابتدائی نمونہ
Observation	Watching and recording what happens without asking questions.	مشاہدہ
Simulation	Computer-based model of a real situation in which variables can be changed.	نقلی ماڈل (سمیولیشن)
Variable	A factor that can be changed or measured.	متغیر
Primary data	Data collected first-hand by the researcher for a specific purpose.	بنیادی ڈیٹا
Secondary data	Data already collected by someone else and reused for a new purpose.	ثانوی ڈیٹا
Research question	The clear question an investigation tries to answer.	تحقیقی سوال
Sample	The group of people who provide the data.	نمونہ
Representative sample	A sample that reflects the mix of the whole group.	نمائندہ نمونہ
Consent	Permission given by a person before their data is collected.	رضامندی / اجازت

Term	Meaning	اردو
Anonymity	Not recording people's names so they cannot be identified.	گمنامی
Ethics	Rules of right and fair behaviour when collecting and using data.	اخلاقیات
Spreadsheet	Software that arranges data in rows and columns and performs calculations.	اسپریدشیت / حسابی جدول
Formula	A spreadsheet instruction starting with = that calculates a result.	فارمولا
Infographic	A visual presentation of information so the main idea is understood quickly.	تصویری معلومات (انفوگرافک)
Report	Structured document presenting data, findings, analysis and conclusions.	رپورٹ
Dashboard	Interactive screen of charts and filters for exploring data.	ڈیش بورڈ
Advanced search operator	Special symbol or word (e.g. " ", filetype:, site:) that narrows search results.	جدید تلاش آپریٹر
Methodology	The plan of methods used to answer a research question.	طریقہ کار
Digital artefact	Final product made with digital technology that answers the research question with evidence.	ڈیجیٹل آرٹیفیکٹ / ڈیجیٹل تخلیق
Recommendation	A practical suggestion based on the findings.	سفارش

Chapter in one look

- **Digital literacy** is the ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly – not just using devices, but finding, evaluating and creating information.
- In the **Information Age**, access to technology is not enough; digital literacy needs technical, cognitive and social abilities.
- **Qualitative data** is descriptive and non-numerical (opinions, reasons, behaviour) and explains why/how.
- **Quantitative data** is numerical, countable or measurable (hours, scores, percentages) and is used for comparing and charting.
- The five **data-collection strategies** are interviews, surveys, prototypes, observation and simulations.
- **Interviews** are planned conversations giving mostly qualitative data; **surveys** ask the same questions to many people, on paper or with Google/Microsoft Forms.
- Good survey questions are clear, short, unbiased and ask about one thing; avoid leading, double-barrelled, vague and overlapping questions.
- A **prototype** is an early version tested for feedback; **observation** records what happens without asking questions; a **simulation** is a computer model used when real experiments are costly, risky or difficult.
- **Primary data** is collected first-hand (high control); **secondary data** was collected by others and reused (limited control).
- Designing a data-collection approach has eight steps: question, data type, method, sample, tools, collect, store with backup, and ethics (consent is mandatory).
- **Spreadsheets** organise and calculate (e.g. `=SUM()`, `=AVERAGE()`, `=COUNTIF()`); **presentations** deliver to an audience; **infographics** give quick visual summaries; **reports** give detailed structure.
- Visuals must always be accurate and based strictly on real data, never exaggerated.
- **Advanced search operators** such as `" "`, `filetype:pdf` and `site:.edu` help find reliable sources efficiently; sources must be evaluated for author, date, reliability and bias.
- **Ethics**: get consent, avoid sensitive information, keep anonymity, keep responses private, and use data only for educational purposes.
- The **digital inquiry project** follows Question → Search → Collect → Analyse → Create in 12 steps.
- A **digital artefact** (report, presentation, infographic or dashboard) answers the research question using evidence and visuals, following a ten-part structure ending in references.

NIGHT BEFORE

One-page recaps

Read each recap, close the booklet, say it aloud. Then stop.

Chapter 1 recap — Computer Systems

- Discrete = counted (stairs). Continuous = measured (ramp). Digital systems use bits 0 and 1; they are reliable, accurate, easy to design and programmable.
- Analog signal = smooth sine wave. Digital signal = square wave HIGH/LOW. Digital data resists noise.
- AND · output 1 only if all inputs 1. OR + output 1 if any input 1. NOT ' inverts. NAND = NOT-AND, NOR = NOT-OR (universal). XOR 1 when inputs differ. XNOR 1 when inputs same.
- Truth table rows = 2^n . SOP = sum of minterms where output is 1. POS = product of maxterms where output is 0.
- K-map: group 1, 2, 4, 8 adjacent 1s (power of 2, wrap-around allowed). Each group drops the variables that change.
- SDLC: requirement → design → implementation → testing → deployment → maintenance. Waterfall = sequential; Agile = short sprints and feedback.
- OSI 7: Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: Please Do Not Throw Sausage Pizza Away. TCP/IP 4: Network Access, Internet, Transport, Application.

Chapter 2 recap — Computational Thinking & Algorithm

- CT pillars: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps. Pseudocode = structured English, not a programming language.
- Bubble Sort: compare neighbours, swap if out of order; largest "bubbles" to the end each pass. $O(n^2)$.
- Selection Sort: find the minimum in the unsorted part and swap it to the front. $O(n^2)$.
- Linear Search: check one by one. Works on unsorted data. $O(n)$.
- Binary Search: list MUST be sorted. Repeatedly test the middle. $O(\log n)$. Trace with low / mid / high.
- Judge an algorithm on correctness, efficiency (time and space) and clarity. Prefer binary search on large sorted lists; linear search on small or unsorted lists.

Chapter 3 recap — Programming Fundamentals

- Python: high-level, interpreted. Variables are names; types int, float, str, bool, list, dict. `input()` returns str.
- Arithmetic `+` `-` `*` `/` `//` `%` `**`. Relational. Logical `and` `or` `not`. Bitwise `&` `|` `^` `~` `<<` `>>`.
- Selection: `if` / `if-else` / `if-elif-else` / nested. Test from the top; first True wins.
- `for` = known count + range. `while` = unknown count; always update the condition. `break` leaves; `continue` skips one.
- Libraries: `import math` / `random` / `datetime`. Third-party via pip: NumPy, Pandas, Matplotlib.
- Syntax = will not run. Runtime = crash + traceback. Logical = runs, wrong answer.

Chapter 4 recap — Data and Analysis

- Data = raw facts. Information = processed data used for decisions. Database = organised related data. DBMS = software that manages it (reduces redundancy, improves security, sharing, integrity).
- Table / relation; row / record / tuple; column / field / attribute. Primary key uniquely identifies a row. Foreign key refers to another table's primary key.
- ER model: entity (rectangle), attribute (oval), relationship (diamond), cardinality 1:1, 1:M, M:N.
- Referential integrity: you cannot orphan a child row. Cascade Update / Cascade Delete copy or remove matching child rows.
- Access objects: tables, forms, queries, reports. Queries filter with criteria, wildcards (`*`, `?`), AND/OR. Totals: Count, Sum, Avg, Group By. Charts visualise the result.

Chapter 5 recap — Application and Impacts of Computing

- IoT = connected devices. Six parts: sensor, actuator, connectivity, processing, cloud, UI. AI = machines that learn and decide. Analytics = collect → store → clean → analyse → visualise.
- Together: IoT (eyes) → Analytics (sort) → AI (brain) → actuator (action).
- Sources: primary (firsthand), secondary (explained by others), tertiary (summaries). No author / date / references → do not trust.
- Assistive tech: mobility, sensory, communication, cognitive. Goal: equal access, independence, inclusion.

Chapter 6 recap — Digital Literacy

- Digital literacy = find, evaluate, create and share information with digital tools, safely.
- Qualitative = words/descriptions. Quantitative = numbers. Strategies: interview, survey, prototype, observation, simulation.
- Primary data you collect; secondary data already exists. Good survey questions are clear, single-idea, unbiased.
- Present with spreadsheets (formulas such as `=AVERAGE()`), slides, infographics and reports. Cite sources. Follow the 12-step inquiry project from question to artefact.

